

Mohammad Haider

Applied mathematician and statistician building software for simulation, robotics, and scientific computing.

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SUMMARY

Simulation-focused software engineer applying statistics and experimental design to robotics and physical systems, especially where scientific questions must become measurable, reproducible experiments.

SELECTED PROJECTS

C-1N + Robotics Test Bench | Python, MuJoCo, Simulation | mhaider.dev/c1n

Jul 2026–Present

- Used Robotics Test Bench experiments to isolate sag in C-1N's original two-joint legs; added an orthogonal coxa degree of freedom / third joint that enabled outward bracing under load.
- Defined standing as a reproducible 10-second acceptance test; validated six contacts, at least 0.23 m of COM-projection margin, at least 0.45 m of torso height, and numerically zero torso angular speed.
- Built a 33-case disturbance suite using 200 ms horizontal force pulses across eight bearings; logged contact, support margin, and torso response, and retained 1x-body-weight failures as an explicit recovery boundary.

PROFESSIONAL EXPERIENCE

University of Virginia | IT Analyst Intermediate, BI Developer | Charlottesville, VA

May 2025–Mar 2026

- Led analytics projects from stakeholder interviews through deployment and maintenance; translated loosely defined operational questions and cross-organizational data needs into SQL-backed Tableau dashboards.
- Built and redesigned the ReUSE Store's environmental and economic impact dashboard as workflows evolved; added decision-oriented views and supported a product that recorded at least 400 views.
- Collaborated on telemetry-informed performance work that improved organization-wide dashboard load times by approximately 20%.
- Reconciled conflicting purchase-order states across financial systems, tracing discrepancies back to their source data and building a dashboard that surfaced deltas and quantified their financial impact.

Microsoft | Software Engineer I | Redmond, WA

Jul 2021–Jul 2024

- Owned an end-to-end migration from SSAS Multidimensional to Tabular; defined parity checks, monitored four weeks of pipeline behavior, and kept the legacy service available until the replacement met cutover criteria.
- Wrote and deployed Azure Data Factory verification scripts that monitored daily pipeline runs and checked data consistency between legacy and Tabular services.
- Validated downstream financial-reporting workflows that showed up to 200x faster filtering and approximately 4x faster Excel loading.
- Built a full-stack C#/.NET prototype that replaced a research-heavy manual configuration process with a guided workflow; explained choices, generated configuration files, and validated output.

Microsoft | Software Engineering Intern | Redmond, WA

Summers 2017, 2018

- Built a C# service that determined each OneDrive user's most recently used items for a product UI feature and validated it in development.

TECHNICAL SKILLS

Languages: Python, SQL, C#, R

Simulation and scientific computing: MuJoCo, NumPy, pandas, Matplotlib, Jupyter

Data and developer workflow: Kusto (KQL), Tableau, Azure Data Factory, Azure Synapse, Git, GitHub Actions, Codex, Claude Code, Hermes

EDUCATION

Virginia Tech | B.S. in Statistics

2021

Coursework: Bayesian Statistics, Experimental Design, Statistical Computing, Mathematical Modeling, Combinatorics, and Proofs